SMART FARM

Real-time greenhouse monitoring, alerts and control

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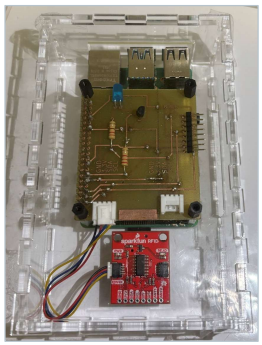
INTRODUCTION

- **Problem** Crop conditions change faster than a farmer can check.
- **Solution** Pi sensor node, cloud database, Android app.
- **Result** Live data, instant alerts, remote control.

1. **SENSE** BME688, VEML7700, OPT4048, RFID on one I²C bus
2. **PROCESS** Raspberry Pi + custom HAT, Python firmware
3. **SYNC** Cloud Firestore over campus Wi-Fi, every 60 s
4. **ACT** Android app: dashboard, alerts, controls

Temperature · Humidity · Light · Fan · Irrigation

ENCLOSURE DESIGN



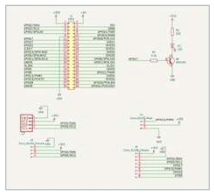
Pi + HAT + RFID reader in the case

- **Laser-cut** clear acrylic
- **Finger-joint** tiles, no glue
- **Open** sides for Qwiic cables

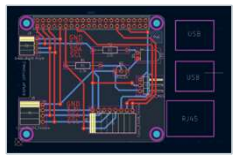
20 W	16 Vrms	32.5 cm
max power	max AC input	max length

PCB ASSEMBLY

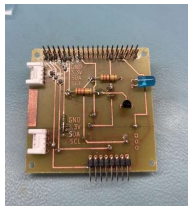
Custom Raspberry Pi HAT KiCad design, hand-soldered



Schematic



Two-layer PCB layout



Assembled HAT

5	60 s	6
I²C devices, one bus	cloud upload interval	OLED status pages

Device	Measures	Address
BME688 (I²C)	Temp, humidity, pressure	0x77
VEML7700 (I²C)	Ambient light	0x10
OPT4048 (I²C)	Colour (Green, Yellow, Red)	0x44
ID-12LA (I²C)	Reading tag ID	0x15
SSD1305 (I²C)	OLED display	0x3C

Python firmware one thread per device

BME688 · VEML7700 · OPT4048 · RFID · OLED

Cloud Firestore
sensor_data / device / readings

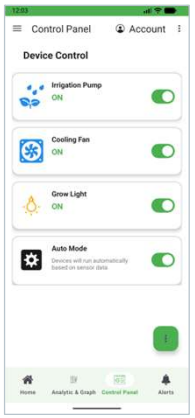
- **Qwiic chain** daisy-chains sensors with no extra wiring
- **Sense HAT removed** to free the I²C bus
- **GPIO17** drives a status LED through a transistor switch

MOBILE APP RESULTS

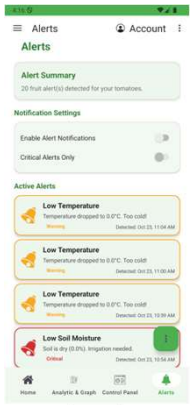
Native Android app Java, Material design, minSdk 31



Dashboard



Control panel



Alerts

History min, max, average graphs

Accounts email or Google, many farms

Settings °C/°F, theme, language

App architecture

- **UI** Home, Analytics, Control Panel, Alerts
- **Controllers** Login, Signup, Account ViewModel
- **Firestore managers** Sensor, Device, Alert, User
- **Firebase** Auth + Cloud Firestore, offline cache

Built with

Java · Material UI · MPAndroidChart · Google Play test

DATABASE

sensor_data
readings: bme688, veml7700, opt4048, rfid

device_controls
pump, fan, light, autoMode

alerts
history with Warning or Critical level

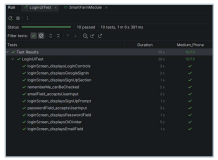
devices
one entry per farm device

users
profile linked to Auth user ID

feedback
ratings + screen_tracking usage

- **Live listeners** push changes to the app
- **Offline cache** keeps the app usable without signal

TESTING AND NEXT STEPS



Espresso login tests: 10 / 10

101
automated tests
JUnit, Robolectric, Espresso

- **Next:** Pi acts on app controls (pump, fan)
- **Next:** fix OPT4048 colour accuracy
- **Next:** open beyond the Play closed test

ACKNOWLEDGEMENTS

Kris Medri, Supervising Faculty

GitHub source code

Google Play closed test

WE ARE HUMBER